

DBMS, models, ER diagrams, keys and normalisation ★ Golden

One college period · 40–45 minutes · Database fundamentals

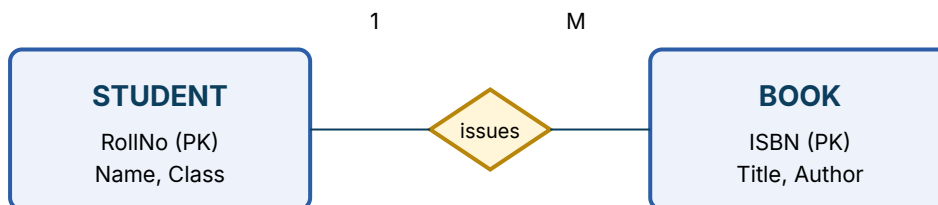
What you will learn

- Differentiate file system and DBMS
- Draw hierarchical, network, relational, object models
- Define PK, FK, candidate, super key; 1NF 2NF 3NF
- Draw a simple ER diagram (library / student-book)

Database. An organised collection of related data. DBMS is software that creates, manages and controls access to the database (MS Access, Oracle, MySQL).

Advantages vs flat files: less redundancy, data integrity, sharing, security, backup, query language. Components: hardware, software, data, procedures, people, DML/DDL (data access language). Users: DBA, application programmer, end user.

Model	Picture
Hierarchical	Tree — one parent, many children (IMS)
Network	Graph — many-to-many with pointers (CODASYL)
Relational	Tables (relations) linked by keys — SQL works here
Object / object-relational	Objects with methods; ORDBMS



Entity Relationship: rectangles = entities, diamond = relationship, ovals = attributes (in full ER).

Term	Meaning
Table / relation	A grid of rows and columns
Record / tuple	A row
Field / attribute	A column
Primary key	Unique identifier, not null
Foreign key	A field that matches the PK of another table
Candidate key	Any field(s) that could be PK
Super key	Any set that uniquely identifies, may have extras
1NF	Atomic cells, no repeating groups
2NF	1NF + non-key fields depend on whole PK
3NF	2NF + no transitive dependence on non-keys

Relationships: 1:1, 1:M, M:M, M:1. Integrity: NOT NULL, PRIMARY KEY, FOREIGN KEY (referential). Three ANSI levels: internal/physical, conceptual/logical, external/view.

HOW BIEK ASKS THIS

Both options 2026: PK vs FK; table, query, report; DBMS components; data models long question; Access data types long question.

اردو میں سمجھیں

ڈیٹا بیس منظم متعلقہ ڈیٹا ہے۔ DBMS سافٹ ویئر (Access) اسے سنبھالتا ہے۔ Primary Key یکتا، Foreign Key دوسری ٹیبل کا لنک۔

Relational ماڈل ٹیبلز اور SQL۔ 1NF 2NF 3NF ڈیٹا دہرانے سے بچاتے ہیں۔

CHECK YOURSELF

1. Row is also called?
2. SQL works on which model?

Answers: 1. Record / tuple · 2. Relational

MCQs

1. A unique key in a table is the:
 - a. Primary key
 - b. Foreign key
 - c. Super key only
 - d. Candidate only — PK is the chosen candidate
2. The primary object in a database is the:
 - a. Table
 - b. Query
 - c. Report
 - d. Form
3. Tree-like database model:

- a. Relational
- b. Hierarchical
- c. Network
- d. Object

4. Columns in a database are:

- a. Tuples
- b. Attributes
- c. Keys only
- d. Tables

5. The model that allows SQL is:

- a. Network
- b. Hierarchical
- c. Relational
- d. Flat

6. DBMS advantage:

- a. data depends on programs
- b. redundancy increases
- c. data is integrated and shared
- d. integrity decreases

Key: 1=a, 2=a, 3=b, 4=b, 5=c, 6=c

Short questions (3 marks each)

1. Differentiate primary key and foreign key.
2. Define table, query, report.
3. Name the major components of DBMS.
4. Who is a DBA? Three responsibilities.
5. Define one-to-many relationship.
6. Write any four advantages of DBMS.
7. Name four objects in MS Access.

Model short answers

1. PK uniquely identifies a row in its own table. FK is a field pointing to a PK in another table to link records.
2. Table stores data. Query asks questions / filters. Report is a printable layout of data.
3. Hardware, software, data, procedures, users, query language.
4. Database Administrator: security, backup, performance, user accounts, recovery.
5. One parent row links to many child rows (one class, many students). Draw 1 — M.
6. Less redundancy, integrity, sharing, security, standard SQL, backup.
7. Tables, Queries, Forms, Reports (Macros, Modules).

Long questions (board Section C)

1. What is a data model? Explain types with diagrams. OR Explain ER modelling and normalisation up to 3NF with a library example.